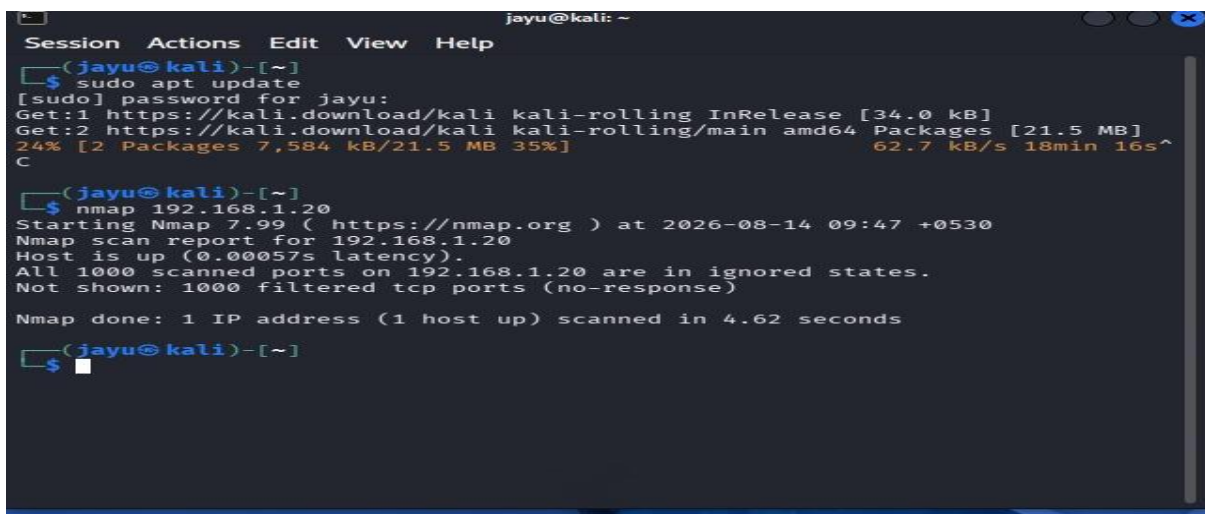


PORT SCANNING CHALLENGE CO3_AT2

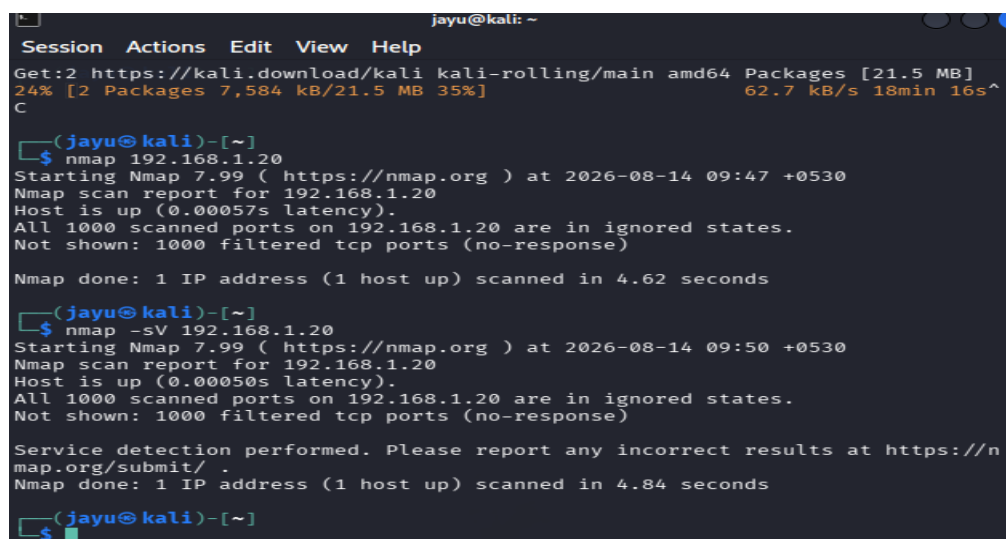
SLOT-B

REGNO.192512417

1. Using Nmap, perform a security assessment on the server 192.168.1.20. Identify all open ports and running services. Analyze the purpose of each discovered service and recommend security measures to reduce the attack surface of the system.

A terminal window titled 'jayu@kali: ~' with a menu bar (Session, Actions, Edit, View, Help). The user runs 'sudo apt update' and then 'nmap 192.168.1.20'. The Nmap scan report shows the host is up with a latency of 0.00057s. All 1000 scanned ports are in ignored states, and 1000 filtered TCP ports were not shown due to no response. The scan took 4.62 seconds.

```
jayu@kali: ~  
Session Actions Edit View Help  
(jayu@kali)-[~]  
$ sudo apt update  
[sudo] password for jayu:  
Get:1 https://kali.download/kali kali-rolling InRelease [34.0 kB]  
Get:2 https://kali.download/kali kali-rolling/main amd64 Packages [21.5 MB]  
24% [2 Packages 7,584 kB/21.5 MB 35%] 62.7 kB/s 18min 16s^  
C  
(jayu@kali)-[~]  
$ nmap 192.168.1.20  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-14 09:47 +0530  
Nmap scan report for 192.168.1.20  
Host is up (0.00057s latency).  
All 1000 scanned ports on 192.168.1.20 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
  
Nmap done: 1 IP address (1 host up) scanned in 4.62 seconds  
(jayu@kali)-[~]  
$
```

A terminal window titled 'jayu@kali: ~' with a menu bar (Session, Actions, Edit, View, Help). The user runs 'nmap -sV 192.168.1.20'. The Nmap scan report shows the host is up with a latency of 0.00050s. All 1000 scanned ports are in ignored states, and 1000 filtered TCP ports were not shown due to no response. The scan took 4.84 seconds. A message prompts the user to report incorrect results at https://nmap.org/submit/.

```
jayu@kali: ~  
Session Actions Edit View Help  
Get:2 https://kali.download/kali kali-rolling/main amd64 Packages [21.5 MB]  
24% [2 Packages 7,584 kB/21.5 MB 35%] 62.7 kB/s 18min 16s^  
C  
(jayu@kali)-[~]  
$ nmap 192.168.1.20  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-14 09:47 +0530  
Nmap scan report for 192.168.1.20  
Host is up (0.00057s latency).  
All 1000 scanned ports on 192.168.1.20 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
  
Nmap done: 1 IP address (1 host up) scanned in 4.62 seconds  
(jayu@kali)-[~]  
$ nmap -sV 192.168.1.20  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-14 09:50 +0530  
Nmap scan report for 192.168.1.20  
Host is up (0.00050s latency).  
All 1000 scanned ports on 192.168.1.20 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 4.84 seconds  
(jayu@kali)-[~]  
$
```

2. Using Nikto, assess the web server hosted on 192.168.1.30. Identify potential web server misconfigurations, outdated components, or exposed files and directories.

Discuss the security risks associated with your findings and suggest appropriate mitigation strategies.

```
jayu@kali: ~  
Session Actions Edit View Help  
Nmap scan report for 192.168.1.30  
Host is up.  
  
PORT      STATE      SERVICE  
80/tcp    filtered  http  
  
Nmap done: 2 IP addresses (2 hosts up) scanned in 0.81 seconds  
  
(jayu@kali)-[~]  
$ nmap -p 80,443 -Pn 192.168.1.30  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-14 11:24 +0530  
Nmap scan report for 192.168.1.30  
Host is up.  
  
PORT      STATE      SERVICE  
80/tcp    filtered  http  
443/tcp   filtered  https  
  
Nmap done: 1 IP address (1 host up) scanned in 3.60 seconds  
  
(jayu@kali)-[~]  
$ nikto -h http://192.168.1.30  
- Nikto v2.6.0  
  
+ [FAIL] Unable to connect to 192.168.1.30:80.  
  
(jayu@kali)-[~]  
$
```

3. Using Netcat (nc), investigate the host 192.168.1.40 by connecting to open ports and collecting available service banners. Analyze the information obtained and explain how attackers could use banner information during the reconnaissance phase of an attack.

```
jayu@kali: ~  
Session Actions Edit View Help  
80/tcp    filtered  http  
443/tcp   filtered  https  
  
Nmap done: 1 IP address (1 host up) scanned in 3.60 seconds  
  
(jayu@kali)-[~]  
$ nikto -h http://192.168.1.30  
- Nikto v2.6.0  
  
+ [FAIL] Unable to connect to 192.168.1.30:80.  
  
(jayu@kali)-[~]  
$ nmap 192.168.1.40  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-14 11:31 +0530  
Nmap scan report for 192.168.1.40  
Host is up (0.0012s latency).  
All 1000 scanned ports on 192.168.1.40 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)  
  
Nmap done: 1 IP address (1 host up) scanned in 4.80 seconds  
  
(jayu@kali)-[~]  
$  
::1          ip6-allnodes      ip6-loopback  
ff02::1      ip6-allrouters    kali  
ff02::2      ip6-localhost      localhost  
  
(jayu@kali)-[~]  
$
```

```
Host is up (0.0012s latency).
All 1000 scanned ports on 192.168.1.40 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 4.80 seconds

(jayu@kali)-[~]
$ nmap 192.168.1.40
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-14 11:31 +0530
Nmap scan report for 192.168.1.40
Host is up (0.0012s latency).
All 1000 scanned ports on 192.168.1.40 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 4.80 seconds

(jayu@kali)-[~]
$ nmap 192.168.1.50
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-14 11:48 +0530
Nmap scan report for 192.168.1.50
Host is up (0.0020s latency).
All 1000 scanned ports on 192.168.1.50 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 4.90 seconds

(jayu@kali)-[~]
$
```

5. Using Netdiscover, perform host discovery on the network segment 192.168.1.1–50. Identify active hosts on the network and analyze how the discovered information could assist an attacker in planning further enumeration or exploitation activities..